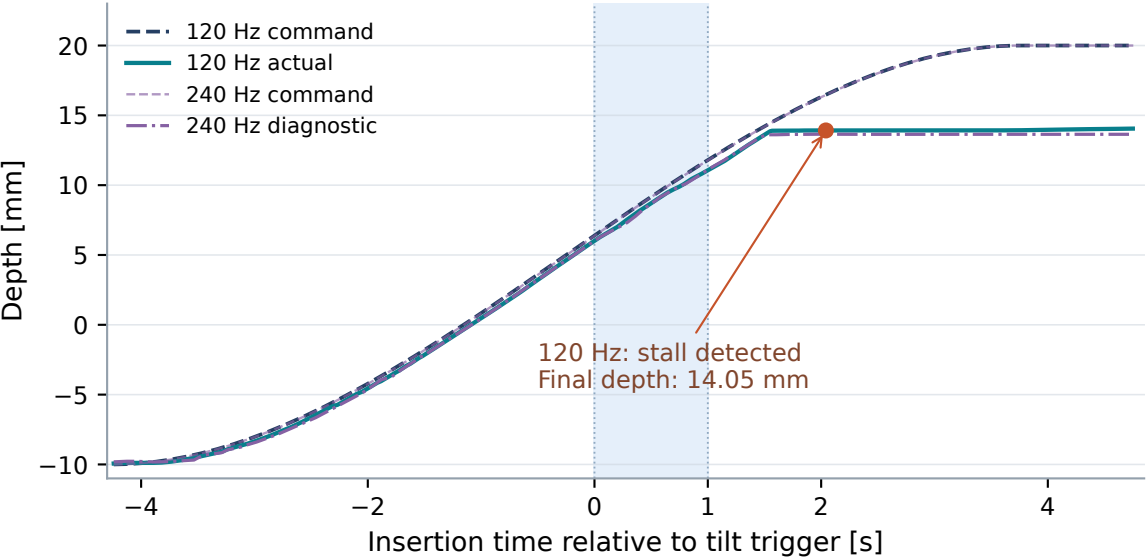


Independent 120 / 240 Hz runs show numerical sensitivity

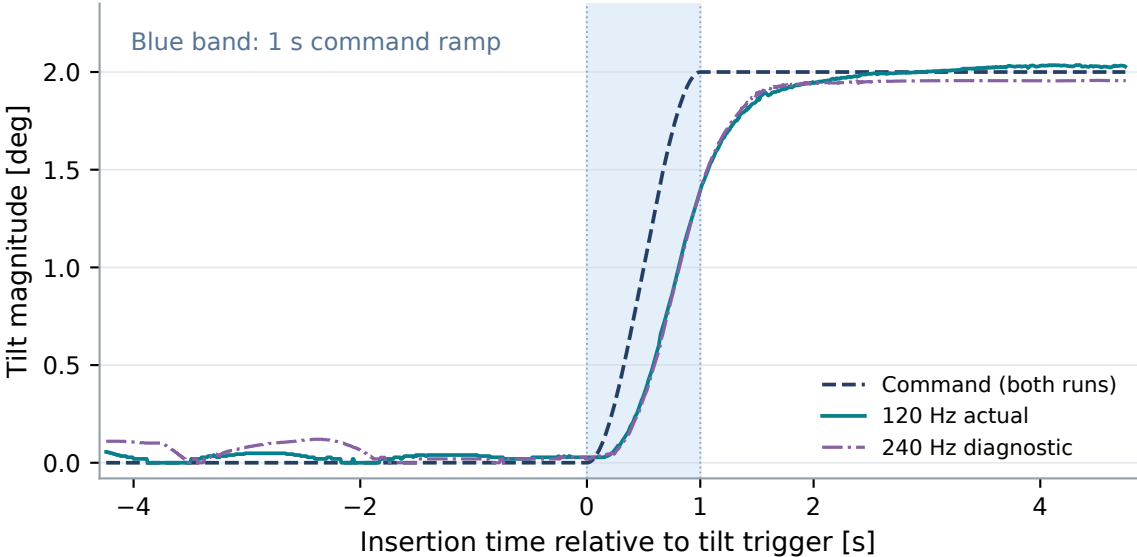
0.2 mm radial clearance · 2° pitch starts at 6 mm actual depth · separate completed references and continuous withdrawals

The retreat peak rises while its 100 ms mean maximum falls; these runs do not establish timestep convergence.

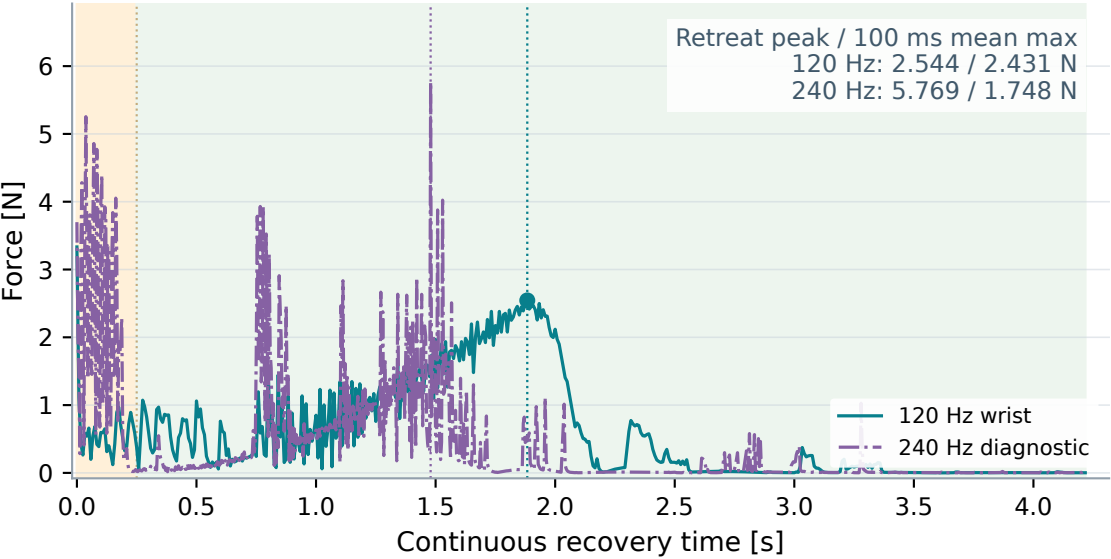
Independent insertion trajectories



Actual tilt under the same nominal command



Raw wrist force: peak and sustained load differ



Amber: stop. Green: withdrawal. Dotted lines mark each retreat peak. Stop peaks: 120 Hz 3.333 N; 240 Hz 5.254 N.

Initial recorded peg positions differ by 0.204 mm. Independent resets and rate-derived controller/filter settings also differ.

Wrist forces are raw reactions. This comparison shows configuration sensitivity; it is not a pure time-step test or a matched recovery-policy pair.

Commanded and actual withdrawal in each run

